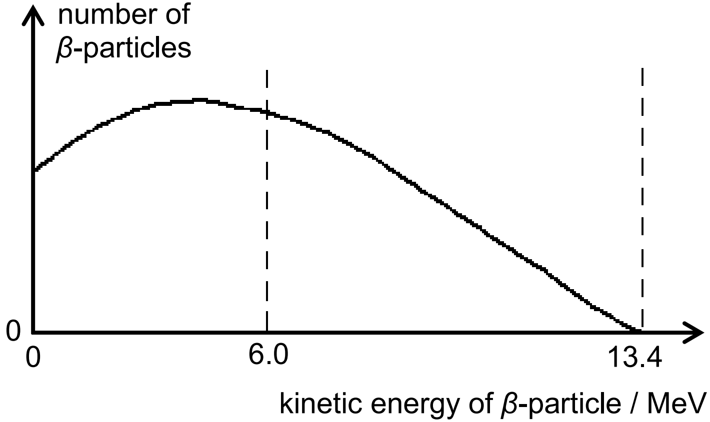
29 The beta spectrum for ^{12}B decay is as shown below.



The kinetic energy of an emitted *β*-particle is 6.0 MeV. What is the approximate energy of the associated neutrino?

A 4.0 MeV **B** 6.0 MeV **C** 7.4 MeV **D** 13.4 MeV

Ans: C

Based on COE and COM, the kinetic energy of the daughter nuclei is negligible hence the total energy released is shared between β particle and neutrino. Since the highest possible KE of β particle is 13.4 MeV, i.e. when neutrino has zero KE, hence the total energy released by the reaction is also 13.4 MeV, and this value is fixed for this particular reaction. Thus, when an emitted β particle has KE of 6.0 MeV, the associated neutrino must have 13.4 – 6.0 = 7.4 MeV of energy.

30 A radioactive source in the laboratory has a half-life of 10 days. The count rate was measured